

The Project Management and Information Technology Context

Different Components in Managing Projects

- **System Approach**
- **Project Life Cycle**
- **Understanding Organizations**

□ Although projects are temporary and is unique, **organizations cannot run projects in isolation and must operate in a broad organizational environment**

□ To effectively handle complex situations, PM need to take a holistic view of a product and understand how it is situated within the larger organization, that is, a **systems view of project management**

□ Take a holistic view of projects and the organizations is called **systems thinking**

□ Use a **systems philosophy, systems analysis and systems management**

□ A **systems philosophy** is an overall model for thinking about things as systems

□ **Systems** are sets of interacting components working within an environment to fulfill some purpose

□ **Systems analysis** is a problem-solving approach that requires defining the scope of the system to be studied, dividing it into its component parts, and then identifying and evaluating its problems, opportunities, constraints and needs, examining alternative solutions for improving the current situation against the entire system

□ **System management** addresses the business, technological, and organizational issues related to each project in order to identify and satisfy key stakeholders and do what is best for the entire organization

3-Sphere Model for Systems Management

- **Business**
- **Organization**
- **Technology**

Project Phases

□ A **project life cycle** is a collection of **project phases**

General Phases:

- **Concept;**
- **Development;**
- **Implementation; and**
- **Close-out.**

□ **The first two phases, concept and development,** focus on planning and are often referred to as project feasibility

□ **The last two phases, implementation and close-out,** focus on delivering the actual work and are often referred to as project acquisition

□ Each phase must be successfully complete before moving on to the next

System Development Life Cycle (SDLC)

□ A **Systems Development Life Cycle (SDLC)** is a framework describing the phases involved in developing and maintaining information systems

□ **Common names of these general phases are information systems planning, analysis, design, implementation, and support.**

Popular Models of a SDLC:

□ **Waterfall model:** well-defined, linear stages of systems development and support

□ **Spiral model:** based on experience with various refinements of the waterfall model

□ **Incremental release model:** provides for progressive development of operational software, with each release providing added capabilities

□ **Prototyping model:** used for developing software prototypes to clarify user requirements for operational software

Management Review

□ A project must successfully pass through each of the project phases in order to continue to the next

□ **Management review** should occur after each phase to **evaluate progress, likely success, continued compatibility with organization goals**

□ These management reviews are called **phase exits** or **kill points**, which plays an important role in keeping projects on track and determine its **continuity, redirection or termination**

□ Having **specific deliverables and kill points** at the **end of project phases** helps managers make better decisions about whether to **proceed, redefine, or kill a project**

□ Improvement in IT project success rates has been due, in part, to an increased ability to know when to cancel failing projects, that is, **"know when to stop beating a dead horse"**

(**Thomas Edison** – his key to success)

□ The systems approach to project management requires that project managers **always view their projects in the context of the larger organization**

(e. g. spend enough time identifying stakeholders, especially oppositions; political context of project in organization)